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The Log of Gravity

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Abstract

Although economists have long been aware of Jensen's inequality, many econometric applications have neglected an important implication of it: the standard practice of interpreting the parameters of log-linearized models estimated by ordinary least squares as elasticities can be highly misleading in the presence of heteroskedasticity. This paper explains why this problem arises and proposes an appropriate estimator. Our criticism to conventional practices and the solution we propose extends to a broad range of economic applications where the equation under study is log-linearized. We develop the argument using one particular illustration, the gravity equation for trade, and apply the proposed technique to provide new estimates of this equation. We find significant differences between estimates obtained with the proposed estimator and those obtained with the traditional method. These discrepancies persist even when the gravity equation takes into account multilateral resistance terms or fixed effects

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1. Introduction

Economists have long been aware that Jensen's inequality implies that $E(\ln y) \neq \ln E(y)$, that is, the expected value of the logarithm of a random variable is different from the logarithm of its expected value. This basic fact, however, has been neglected in many econometric applications. Indeed, one important implication of Jensen's inequality is that the standard practice of interpreting the parameters of log-linearized models estimated by ordinary least squares (OLS) as elasticities can be highly misleading in the presence of heteroskedasticity.

Although many authors have addressed the problem of obtaining consistent estimates of the conditional mean of the dependent variable when the model is estimated in the log-linear form (see, for example, Goldberger, 1968; Manning and Mullahy, 2001), we were unable to find any reference in the literature to the potential bias of the elasticities estimated using the log-linear model.

In this paper we use the gravity equation for trade as a particular illustration of how the bias arises and propose an appropriate estimator. We argue that the gravity equation, and, more generally, constant-elasticity models, should be estimated in their multiplicative form and propose a simple pseudo-maximum likelihood (PML) estimation technique. Besides being consistent in the presence of heteroskedasticity, this method also provides a natural way to deal with zero values of the dependent variable.

Using Monte Carlo simulations, we compare the performance of our estimator with that of OLS (in the log-linear specification). The results are striking. In the presence of heteroskedasticity, estimates obtained using log-linearized models are severely biased, distorting the interpretation of the model. These biases might be critical for the comparative assessment of competing economic theories, as well as for the evaluation of the effects of different policies. In contrast, our method is robust to the different patterns of heteroskedasticity considered in the simulations.

We next use the proposed method to provide new estimates of the gravity equation in cross sectional data. Using standard tests, we show that heteroskedasticity is indeed a

severe problem, both in the traditional gravity equation introduced by Tinbergen (1962), and in a gravity equation that takes into account multilateral resistance terms or fixed effects, as suggested by Anderson and van Wincoop (2003). We then compare the estimates obtained with the proposed PML estimator with those generated by OLS in the log-linear specification, using both the traditional and the fixed-effects gravity equations.

Our estimation method paints a very different picture of the determinants of international trade. In the traditional gravity equation, the coefficients on GDP are not, as generally estimated, close to one. Instead, they are significantly smaller, which might help reconcile the gravity equation with the observation that the trade-to-GDP ratio decreases with total GDP (or, in other words, that smaller countries tend to be more open to international trade). In addition, OLS greatly exaggerates the roles of colonial ties, geographical proximity, and bilateral trade agreements.

Using the Anderson-van Wincoop (2003) gravity equation, we find that OLS yields significantly larger effects for geographical distance. The estimated elasticity obtained from the log-linearized equation is almost twice as large as that predicted by PML. OLS also predicts a large role for common colonial ties, implying that sharing a common colonial history practically doubles bilateral trade. In contrast, the proposed PML estimator leads to a statistically and economically insignificant effect.

The general message is that, even controlling for fixed effects, the presence of heteroskedasticity can generate strikingly different estimates when the gravity equation is log-linearized, rather than estimated in levels. In other words, Jensen's inequality is quantitatively and qualitatively important in the estimation of gravity equations. This suggests that inferences drawn on log-linearized regressions can produce misleading conclusions.

Despite the focus on the gravity equation, our criticism to the conventional practice and the solution we propose extends to a broad range of economic applications where the equations under study are log-linearized, or, more generally, transformed by a non-linear function. A short list of examples includes the estimation of Mincerian equations